

Technology Stack

Date	29 September 2026
Team ID	6ab2a427dbd693aadc5a7d4b
Project Name	Student Enrollment & Course Management System (SECMS)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js with HTML5, CSS3, Bootstrap 5	Responsive design, component reusability, fast rendering for student dashboard, SPA for smooth experience
2	Backend / Server-Side	Node.js with Express.js	Fast processing, handles 10k+ concurrent requests, RESTful API, team expertise, easy payment integration
3	Database / Data Storage	MySQL + MongoDB	MySQL for relational Student-Course-Payment mapping with ACID, MongoDB for unstructured docs like certificates

4	Cloud / Hosting / Deployment	AWS EC2 + S3 + Docker	Easy scalability during admission peak, cost-effective, S3 for storage, Docker container, auto-scaling
5	Version Control & CI/CD	GitHub + Jenkins + GitHub Actions	Seamless collaboration, version tracking, automated testing & deployment to AWS
6	Third-Party APIs / Other Tools	Razorpay, SendGrid, Twilio, Google Auth	Razorpay for secure fee payment, SendGrid/Twilio for OTP & notifications, Google Auth for SSO

